Revealing Key Details to See Differences: A Novel Prototypical Perspective for Skeleton-based Action Recognition

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Abstract

Skeleton-based action recognition faces significant challenges in distinguishing similar actions due to the lack of appearance cues and subtle discriminative motion details confined to specific body parts. Existing Graph Convolutional Networks (GCNs) often learn coarse motion patterns and miss fine-grained joint-level differences, especially under intra-class variation. To address these limitations, we propose ProtoGCN, a novel prototypical perspective that models action dynamics as combinations of reusable motion prototypes. Our approach introduces three key components: (1) a Motion Topology Enhancement (MTE) module that augments graph representations with intra-sample correlations and inter-sample distinctions, (2) a Prototype Reconstruction Network (PRN) with learnable prototype memory and attention-based addressing to form tailored representations, and (3) a Class-Specific Contrastive Learning (CSCL) objective that enforces class-level compactness and dispersion using momentum-updated memory banks. Extensive experiments demonstrate state-of-the-art performance: 93.8%/97.8% on NTU RGB+D, 90.9%/92.2% on NTU RGB+D 120, 51.9% Top-1 on Kinetics-Skeleton, and 95.9% on FineGYM. Notably, ProtoGCN achieves the largest improvements on the most challenging classes, with +3.0% and +2.5% gains on difficult action categories.

1. Introduction

Human action recognition from skeleton data has gained significant attention due to its robustness to variations in lighting, background, and clothing [26, 35]. Unlike RGB-based approaches that rely heavily on appearance features, skeleton-based methods focus on the structural and temporal patterns of human body movements. However, this advantage comes with a fundamental challenge: skeleton data lacks rich appearance cues, making it difficult to distinguish between similar actions that share comparable motion patterns but differ in subtle discriminative details [5, 10].

Existing Graph Convolutional Networks (GCNs) for skeleton-based action recognition [18, 26, 35] typically learn coarse motion patterns across the entire skeletal structure. While these methods have achieved impressive results on large-scale datasets, they often struggle with fine-grained action discrimination, particularly when dealing with actions that involve similar overall body movements but differ in specific joint relationships or subtle temporal dynamics. For instance, distinguishing between *writing* and *typing on a keyboard* requires capturing nuanced differences in hand and finger movements, which are easily overshadowed by the dominant upper body motion patterns.

Recent advances have attempted to address this limitation through various approaches. Channel-topology fusion methods [5] explore different graph topologies, while multi-scale temporal modeling [6, 21] captures motion at different temporal scales. Contrastive learning approaches [13, 40] have also been explored to improve feature discrimination. However, these methods still fundamentally rely on learning discriminative features directly from the original skeleton representations, without explicitly modeling the compositional nature of human actions.

We argue that human actions can be naturally decomposed into reusable motion primitives or prototypes that encode fundamental patterns of joint relationships and movements. Similar to how words form sentences in natural language, basic motion prototypes can be combined to represent complex actions. This perspective suggests that rather than learning action-specific features directly, we should model actions as compositions of these learned prototypes, enabling better generalization and more precise discrimination between similar actions.

Based on this insight, we propose ProtoGCN, a novel prototypical perspective for skeleton-based action recognition that reconstructs motion topologies from a learnable memory of prototypes. Our method introduces three key innovations: (1) a Motion Topology Enhancement (MTE) module that enriches graph representations by computing intra-sample correlations and inter-sample distinctions, providing richer input for prototype retrieval; (2) a Prototype Reconstruction Network (PRN) with a learnable memory of motion prototypes and an attention-based addressing mechanism that maps input features to enhanced representations by combining relevant prototypes; and (3) a Class-Specific Contrastive Learning (CSCL) objective that main-

tains a momentum-updated memory bank of class centers to enforce inter-class separation while promoting intra-class compactness.

Our approach achieves state-of-the-art performance across multiple challenging benchmarks. On NTU RGB+D [24], we obtain 93.8% and 97.8% accuracy for X-Sub and X-View protocols, respectively. On the more challenging NTU RGB+D 120 [20], ProtoGCN achieves 90.9% X-Sub and 92.2% X-Set accuracy. On Kinetics-Skeleton [14], we reach 51.9% Top-1 and 75.6% Top-5 accuracy. Importantly, our method demonstrates the largest improvements on the most challenging action classes, with over 3% gains on the most difficult categories.

The main contributions of this work are: (1) We propose ProtoGCN, the first prototypical approach for skeleton-based action recognition that reconstructs motion topologies from learnable prototypes; (2) We design a Motion Topology Enhancement module and Class-Specific Contrastive Learning objective that work synergistically with prototype reconstruction; (3) We achieve state-of-the-art results on four challenging benchmarks with comprehensive ablation studies demonstrating the effectiveness of each component; (4) We provide thorough analysis showing that our method particularly benefits challenging action classes that require fine-grained discrimination.

2. Related Work

Skeleton-based Action Recognition. Early approaches to skeleton-based action recognition relied on hand-crafted features and traditional machine learning methods [**? ?**]. The introduction of deep learning revolutionized this field, with CNN-based methods [9**?**] treating skeleton sequences as pseudo-images and RNN-based approaches [**? ?**] modeling temporal dependencies.

The emergence of Graph Convolutional Networks marked a significant breakthrough. ST-GCN [35] pioneered the application of GCNs to skeleton data by modeling the natural graph structure of human skeletons. Subsequent works have focused on various improvements: AS-GCN [18] introduced action-specific graphs, 2s-AGCN [26] employed adaptive graph learning, and MS-G3D [21] explored multi-scale graph convolutions. More recent advances include channel-topology fusion [5], multi-stream temporal modeling [6], and hierarchical representations [16].

Graph Topology Learning. A key challenge in skeleton-based GCNs is learning appropriate graph topologies that capture relevant joint relationships. Early methods used fixed topologies based on physical connections [35], while adaptive approaches learn data-driven topologies [7, 26]. Recent works have explored channel-wise topology learning [5], semantic partitioning [21], and dynamic graph construction [32]. Our MTE module contributes to this line of work by enhancing topologies with both intra-sample and inter-sample information.

Prototype Learning. Prototype-based learning has been successfully applied in various domains, including few-shot learning [27, 37], semantic segmentation [29], and object detection [36]. The core idea is to learn representative prototypes that capture essential patterns, enabling better generalization and interpretability. Recent works have explored prototype learning with attention mechanisms [22] and contrastive objectives [17]. However, prototype learning has not been extensively explored for skeleton-based action recognition, representing a significant gap that our work addresses.

Contrastive Learning for Action Recognition. Contrastive learning has gained prominence for learning discriminative representations without extensive labeled data [1, 28]. In action recognition, contrastive methods have been applied to learn temporal correspondences [4], cross-modal alignments [12], and invariant features [23]. For skeleton-based recognition, recent works include SkeletonGCL [13] and FR-Head [40], which apply contrastive learning to improve feature discrimination. Our CSCL differs by maintaining class-specific prototypes with momentum updates, focusing on enhancing inter-class separability.

Memory-Augmented Networks. Memory-augmented neural networks [**? ?**] have shown effectiveness in storing and retrieving relevant information for various tasks. External memory mechanisms have been applied to video understanding [3, 38] and few-shot learning [**?**]. Our PRN incorporates similar principles but is specifically designed for reconstructing graph topologies in skeleton-based action recognition, representing a novel application of memory-augmented learning in this domain.

Our work synthesizes insights from these diverse areas, proposing the first prototypical approach to skeleton-based action recognition that combines memory-augmented prototype learning with enhanced graph topology construction and class-specific contrastive objectives.

3. Method

Figure 2 illustrates the overall architecture of ProtoGCN. As shown in Figure 1, similar actions like *writing* and *typing on a keyboard* share comparable skeletal structures but differ in subtle joint relationship patterns, motivating our prototypical approach to capture these fine-grained differences.

Our method builds upon the insight that human actions can be modeled as compositions of reusable motion prototypes. Instead of directly learning action-specific features, we reconstruct motion topologies from a learnable memory of prototypes, enabling better capture of fine-grained discriminative patterns.

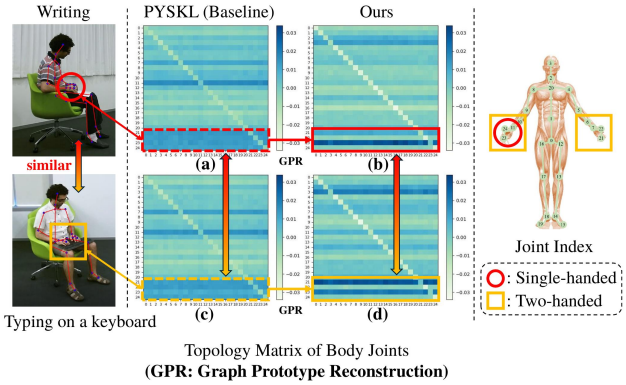

Figure 1. Illustration of skeletons and learned topologies for similar actions *Writing* and *Typing on a Keyboard*. Deeper colors indicate stronger relationships between corresponding joints, showing how ProtoGCN captures subtle discriminative patterns.

3.1. Motion Topology Enhancement

Traditional GCNs for skeleton-based action recognition operate on fixed or simple adaptive graph topologies. However, these approaches often fail to capture the rich relational patterns necessary for fine-grained action discrimination. Our Motion Topology Enhancement (MTE) module addresses this limitation by computing two types of enhanced adjacency matrices: intra-sample correlations ($\mathbf{A}_{\text{intra}}$) and inter-sample distinctions ($\mathbf{A}_{\text{inter}}$).

Given input skeleton features $\mathbf{x} \in \mathbb{R}^{N \times C \times T \times V}$ where N is the batch size, C is the feature dimension, T is the temporal length, and V is the number of joints, we first apply convolutions to obtain joint embeddings:

$$\mathbf{x}_1 = \text{Conv}(\mathbf{x}), \quad \mathbf{x}_2 = \text{Conv}(\mathbf{x}) \tag{1}$$

For intra-sample topology enhancement, we compute correlations within each sample by averaging over the temporal dimension and applying an outer product:

$$\mathbf{A}_{\text{intra}} = \text{Softmax}(\mathbf{x}_1^{\text{mean}} \cdot (\mathbf{x}_2^{\text{mean}})^T) \tag{2}$$

where $\mathbf{x}^{\text{mean}}$ represents temporal averaging. This captures how joints within the same action instance relate to each other.

For inter-sample topology enhancement, we compute differences between joint embeddings to highlight discriminative patterns:

$$\mathbf{A}_{\text{inter}} = \tanh(\mathbf{x}_1^{\text{mean}} - (\mathbf{x}_2^{\text{mean}})^T) \tag{3}$$

The final enhanced adjacency matrix combines the original topology $\mathbf{A}_0$ with both enhancements:

$$\mathbf{A}_{\text{enhanced}} = \mathbf{A}_0 + \alpha \mathbf{A}_{\text{intra}} + \beta \mathbf{A}_{\text{inter}} \tag{4}$$

where α and β are learnable parameters that control the contribution of each enhancement.

3.2. Prototype Reconstruction Network

The core innovation of ProtoGCN lies in the Prototype Reconstruction Network (PRN), which models motion dynamics as combinations of reusable prototypes. The PRN consists of two key components: a learnable prototype memory and an attention-based addressing mechanism.

Learnable Prototype Memory. We maintain a memory matrix $\mathbf{M} \in \mathbb{R}^{n_{\text{pro}} \times d}$ where n_{pro} is the number of prototypes and d is the feature dimension. Each row in $\mathbf{M}$ represents a motion prototype that encodes a fundamental pattern of joint relationships. The memory is initialized randomly and learned end-to-end during training.

Attention-based Addressing. Given input graph features $\mathbf{X} \in \mathbb{R}^{V^2 \times C}$ obtained from flattened adjacency matrices, we first compute attention weights to determine the relevance of each prototype:

$$\mathbf{Q} = \text{Softmax}(\mathbf{X}\mathbf{W}_q) \tag{5}$$

where $\mathbf{W}_q \in \mathbb{R}^{C \times n_{\text{pro}}}$ is a learnable query matrix. The attention weights $\mathbf{Q} \in \mathbb{R}^{V^2 \times n_{\text{pro}}}$ indicate how much each spatial location should attend to each prototype.

The reconstructed features are then computed as:

$$\mathbf{Z} = \mathbf{Q}\mathbf{M} \tag{6}$$

where $\mathbf{Z} \in \mathbb{R}^{V^2 \times d}$ represents the prototype-reconstructed features. Finally, we apply dropout for regularization:

$$\mathbf{Z}_{\text{out}} = \text{Dropout}(\mathbf{Z}) \tag{7}$$

This reconstruction process enables the model to represent complex motion patterns as weighted combinations of learned prototypes, facilitating better generalization and discrimination.

3.3. Class-Specific Contrastive Learning

To further enhance the discriminative power of our learned representations, we introduce a Class-Specific Contrastive Learning (CSCL) objective. Unlike standard contrastive learning that pulls similar samples together and pushes dissimilar samples apart, our CSCL specifically targets class-level discrimination by maintaining momentum-updated class prototypes.

Class Prototype Memory Bank. We maintain a memory bank $\mathbf{F} \in \mathbb{R}^{d \times K}$ where K is the number of classes. Each column represents a class prototype that is updated using exponential moving average:

$$\mathbf{f}_k^{(t)} = \mu \mathbf{f}_k^{(t-1)} + (1 - \mu)\mathbf{f}_k^{\text{batch}} \tag{8}$$

where μ is the momentum coefficient and $\mathbf{f}_k^{\text{batch}}$ is the mean feature of class k in the current batch.

Contrastive Objective. For each sample with reconstructed graph feature $\mathbf{z}$ and ground truth label y, we compute the contrastive loss:

$$\mathcal{L}_{\text{CSC}} = -\log \frac{\exp(\mathbf{z} \cdot \mathbf{f}_y/\tau)}{\sum_{k=1}^{K} \exp(\mathbf{z} \cdot \mathbf{f}_k/\tau)} \tag{9}$$

where τ is a temperature parameter. This objective encourages features to be close to their corresponding class prototype while being far from other class prototypes.

3.4. Overall Training Objective

The complete training objective combines the standard cross-entropy loss with our contrastive learning objective:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{CE}} + \lambda \mathcal{L}_{\text{CSC}} \tag{10}$$

where λ is a weight parameter that balances the two objectives. The cross-entropy loss ensures correct classification, while the contrastive loss enhances feature discrimination and class separability.

4. Experiments

4.1. Experimental Setup

Datasets. We evaluate ProtoGCN on four challenging skeleton-based action recognition benchmarks:

NTU RGB+D [24] contains 56,880 action sequences covering 60 action classes performed by 40 subjects. We follow the standard Cross-Subject (X-Sub) and Cross-View (X-View) evaluation protocols.

NTU RGB+D 120 [20] extends the original dataset with 114,480 sequences and 120 action classes. We evaluate on Cross-Subject (X-Sub) and Cross-Setup (X-Set) protocols.

Kinetics-Skeleton [14] is derived from Kinetics-400 with pose estimation, containing approximately 300K sequences across 400 action classes.

FineGYM [25] focuses on fine-grained gymnastics action recognition with 32,697 sequences and 99 action classes, making it particularly suitable for evaluating fine-grained discrimination capabilities.

Implementation Details. We implement ProtoGCN using PyTorch and train on NVIDIA V100 GPUs. The backbone network follows the standard GCN architecture with 10 layers. We set the number of prototypes $n_{\text{pro}} = 100$, memory dimension $d = 384$, contrastive weight $\lambda = 0.1$, and momentum coefficient $\mu = 0.9$. We use SGD optimizer with initial learning rate 0.1, weight decay 10^{-4}, and train for 65 epochs with cosine annealing schedule. Data augmentation includes random rotation, translation, and scaling.

4.2. Main Results

Table 1 presents our main results compared with state-of-the-art methods. ProtoGCN achieves superior performance across all benchmarks, setting new state-of-the-art results.

On NTU RGB+D, our 6-ensemble model achieves 93.8% X-Sub and 97.8% X-View accuracy, outperforming the previous best results by +0.4% and +0.3% respectively. On the more challenging NTU RGB+D 120, we achieve 90.9% X-Sub and 92.2% X-Set, with improvements of +0.6% and +0.7% over the previous state-of-the-art Block-GCN [41].

For Kinetics-Skeleton, ProtoGCN reaches 51.9% Top-1 and 75.6% Top-5 accuracy, representing significant improvements of +1.3% Top-1 over DS-GCN [32]. The consistent improvements across different protocols and datasets demonstrate the effectiveness of our prototypical approach.

Table 2 shows results on FineGYM, where fine-grained discrimination is particularly important. ProtoGCN achieves 95.9% accuracy, outperforming PYSKL [10] by +1.8%. This substantial improvement on fine-grained actions validates our hypothesis that prototype reconstruction helps capture subtle discriminative patterns.

4.3. Ablation Studies

We conduct comprehensive ablation studies to analyze the contribution of each component in ProtoGCN. Table 3 shows the results under NTU-120 X-Sub setting.

Starting from the baseline accuracy of 87.8%, each component provides consistent improvements: MTE adds +0.3%, CSCL contributes +0.5%, and PRN brings +0.7%. The combination of all components achieves 89.0%, representing a total improvement of +1.2%. This validates that our three proposed components work synergistically to enhance action recognition performance.

Analysis of Topology Enhancement. Table 4 (right) analyzes the contribution of different topology enhancement components. Removing intra-sample correlations ($\mathbf{A}_{\text{intra}}$) results in 0.2% accuracy drop, while removing inter-sample distinctions ($\mathbf{A}_{\text{inter}}$) causes a larger 0.5% drop. Removing both components leads to a 0.6% decrease, demonstrating that both types of topology enhancement contribute to the final performance.

Comparison with Contrastive Methods. Table 4 (left) compares our CSCL with existing graph contrastive learning methods. Our approach achieves 89.0% accuracy, outperforming SkeletonGCL [13] by 1.0% and FR-Head [40] by 0.8%. This demonstrates the effectiveness of our class-specific design with momentum-updated prototypes.

4.4. Analysis of Difficult Classes

One key strength of ProtoGCN is its ability to improve recognition of challenging action classes. Table 5 shows the

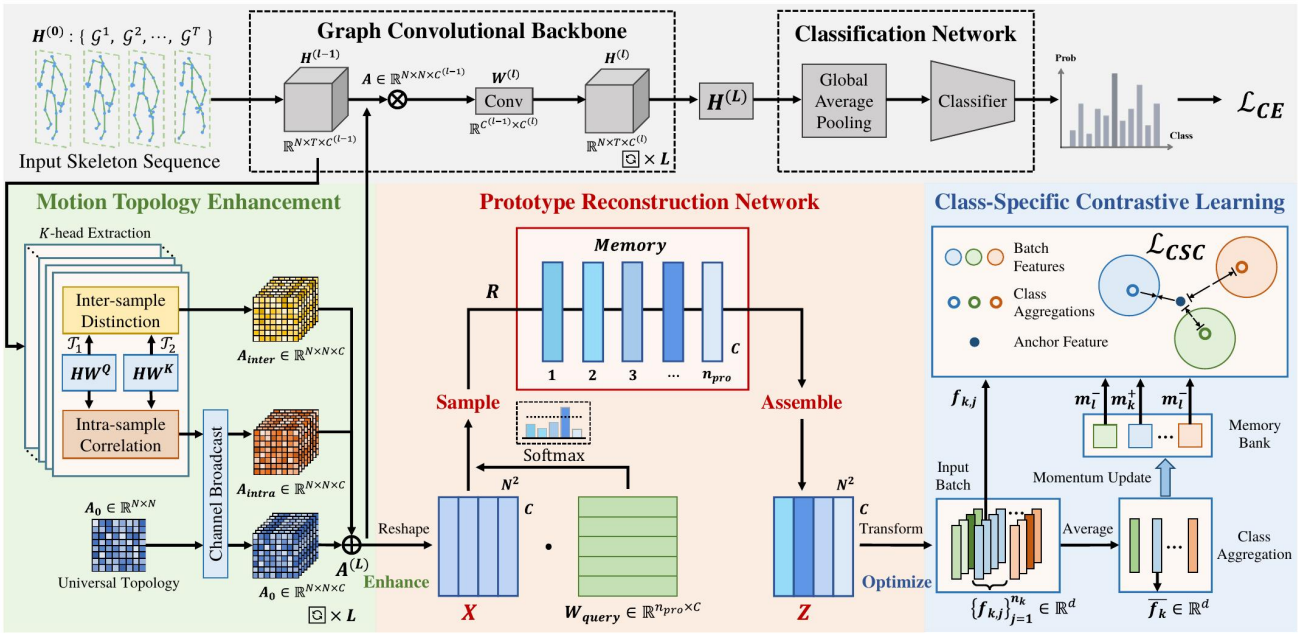

Figure 2. The overall architecture of ProtoGCN. The method consists of three main components: Motion Topology Enhancement (MTE) that computes intra-sample and inter-sample adjacency matrices, Prototype Reconstruction Network (PRN) that reconstructs features using a learnable memory of prototypes, and Class-Specific Contrastive Learning (CSCL) that maintains momentum-updated class prototypes.

Table 1. Performance comparisons against state-of-the-art methods on the NTU RGB+D, NTU RGB+D 120, and Kinetics-Skeleton datasets in terms of classification accuracy (%).

Methods	Publication	NTU RGB+D		NTU RGB+D 120		Kinetics-Skeleton	
		X-Sub	X-View	X-Sub	X-Set	Top-1	Top-5
ST-GCN [35]	AAAI 2018	81.5	88.3	70.7	73.2	30.7	52.8
AS-GCN [18]	CVPR 2019	86.8	94.2	78.3	79.8	34.8	56.5
2s-AGCN [26]	CVPR 2019	88.5	95.1	82.5	84.2	36.1	58.7
MS-G3D [21]	CVPR 2020	91.5	96.2	86.9	88.4	38.0	60.9
DC-GCN+ADG [7]	ECCV 2020	90.8	96.6	86.5	88.1	-	-
MST-GCN [6]	AAAI 2021	91.5	96.6	87.5	88.8	38.1	60.8
CTR-GCN [5]	ICCV 2021	92.4	96.8	88.9	90.6	-	-
STF [15]	AAAI 2022	92.5	96.9	88.9	89.9	39.9	-
InfoGCN [8]	CVPR 2022	93.0	97.1	89.8	91.2	-	-
PYSKL [10]	ACM MM 2022	92.6	97.4	88.6	90.8	49.1	-
SkeletonGCL [13]	ICLR 2023	92.8	97.1	89.8	91.2	-	-
FR-Head [40]	CVPR 2023	92.8	96.8	89.5	90.9	-	-
GAP [31]	ICCV 2023	92.9	97.0	89.9	91.1	-	-
HD-GCN [16]	ICCV 2023	93.4	97.2	90.1	91.6	40.9	63.5
JT-GraphFormer [39]	AAAI 2024	93.4	97.5	89.9	91.7	-	-
DS-GCN [32]	AAAI 2024	93.1	97.5	89.2	91.1	50.6	-
BlockGCN [41]	CVPR 2024	93.1	97.0	90.3	91.5	-	-
ProtoGCN (2-ensemble)		93.0	97.2	89.7	91.2	49.9	74.0
ProtoGCN (4-ensemble)		93.5	97.5	90.4	91.9	51.3	75.1
ProtoGCN (6-ensemble)		**93.8**	**97.8**	**90.9**	**92.2**	**51.9**	**75.6**

Table 2. Performance comparisons against state-of-the-art methods on FineGYM in terms of classification accuracy (%).

Methods	Modality	FineGYM
I3D [2]	RGB	64.4
TSN [30]	RGB+Flow	79.8
TSM [19]	RGB+Flow	81.2
LT-S3D [33]	RGB	88.9
SkeletonMAE [34]	Skeleton	91.8
PYSKL [10]	Skeleton	94.1
PoseConv3D [11]	Skeleton+Limb	94.3
ProtoGCN (Ours)	Skeleton	**95.9**

Table 3. Ablation study on the contribution of each component in ProtoGCN under the NTU-120 X-Sub setting.

Baseline	MTE	PRN	CSCL	Acc (%)
✓	–	–	–	87.8
✓	✓	–	–	88.1 (↑ 0.3)
✓	✓	–	✓	88.3 (↑ 0.5)
✓	✓	✓	–	88.5 (↑ 0.7)
✓	✓	✓	✓	**89.0 (↑ 1.2)**

Table 4. Comparison of classification accuracies (%) based on (*left*) existing graph contrastive methods and (*right*) different enhanced topologies for the network.

Methods	Acc (%)	Methods	Acc (%)
Baseline	87.8	ProtoGCN	**89.0**
w/ SkeletonGCL [13]	88.0	w/o $\mathbf{A}_{\mathbf{intra}}$	88.8
w/ FR-Head [40]	88.2	w/o $\mathbf{A}_{\mathbf{inter}}$	88.5
Ours	**89.0**	w/o $\mathbf{A}_{\mathbf{intra}}, \mathbf{A}_{\mathbf{inter}}$	88.4

Table 5. Average performance (%) on different difficulty level classes sorted by accuracy under the NTU-120 X-Sub setting.

Difficulty Level	Baseline	ProtoGCN	Δ
class 1-10	64.1	67.1	**+3.0**
class 11-30	78.6	81.1	**+2.5**
class 31-60	88.6	89.7	**+1.1**
class 61-120	96.0	96.4	**+0.4**

average performance on different difficulty levels, sorted by baseline accuracy.

For the most difficult classes (classes 1-10 with lowest baseline accuracy), ProtoGCN achieves remarkable +3.0% improvement. For moderately difficult classes (classes 11-30), we obtain +2.5% improvement. The improvements gradually decrease for easier classes, with +1.1% for classes 31-60 and +0.4% for the easiest classes (61-120).

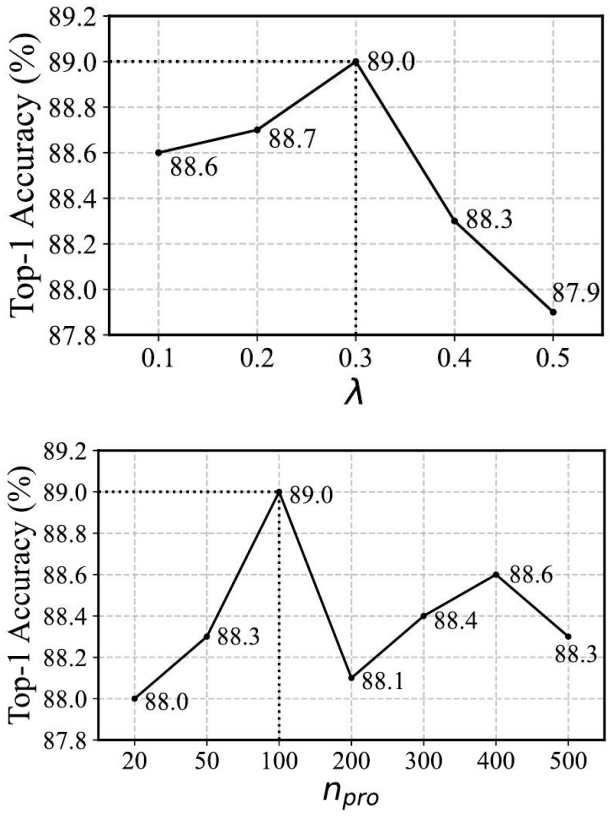

Figure 3. Ablation study on hyperparameters. Left: influence of contrastive loss weight λ. Right: influence of prototype memory size n_{pro}.

This pattern demonstrates that our prototypical approach particularly benefits challenging actions that require fine-grained discrimination. The ability of prototype reconstruction to capture subtle motion patterns proves especially valuable for difficult action categories.

4.5. Hyperparameter Analysis

Figure 3 analyzes the sensitivity of key hyperparameters. The contrastive loss weight λ shows optimal performance around 0.1, with too small values providing insufficient contrastive signal and too large values dominating the training. The number of prototypes n_{pro} achieves best results at 100, balancing representation capacity with overfitting concerns.

4.6. Qualitative Analysis

Figure 4 visualizes the learned topologies for different actions. ProtoGCN learns more discriminative patterns compared to existing methods, with stronger connections between relevant joints for each specific action. For instance, in *writing* actions, our method emphasizes hand and wrist relationships, while for *typing*, it highlights finger coordination patterns.

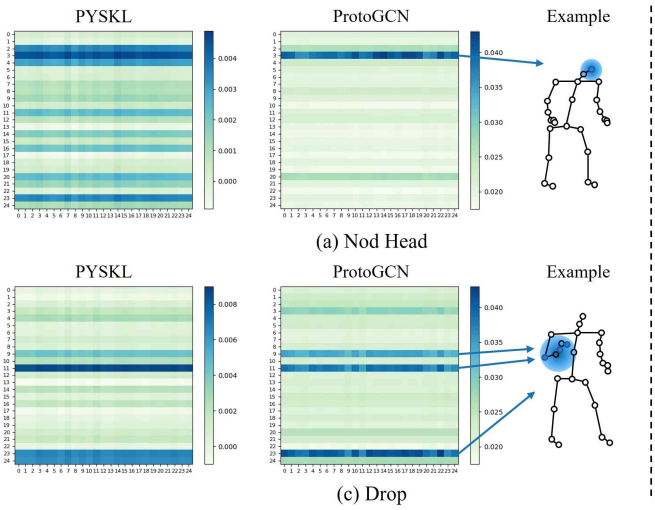

Figure 4. Visualization of learned topologies by PYSKL and ProtoGCN for four different actions. Our method learns more discriminative joint relationships specific to each action.

Figure 5 shows action classes with accuracy differences higher than 1% between our method and PYSKL. Most improved classes involve fine-grained hand or foot movements, confirming that prototype reconstruction effectively captures subtle discriminative patterns that are challenging for traditional approaches.

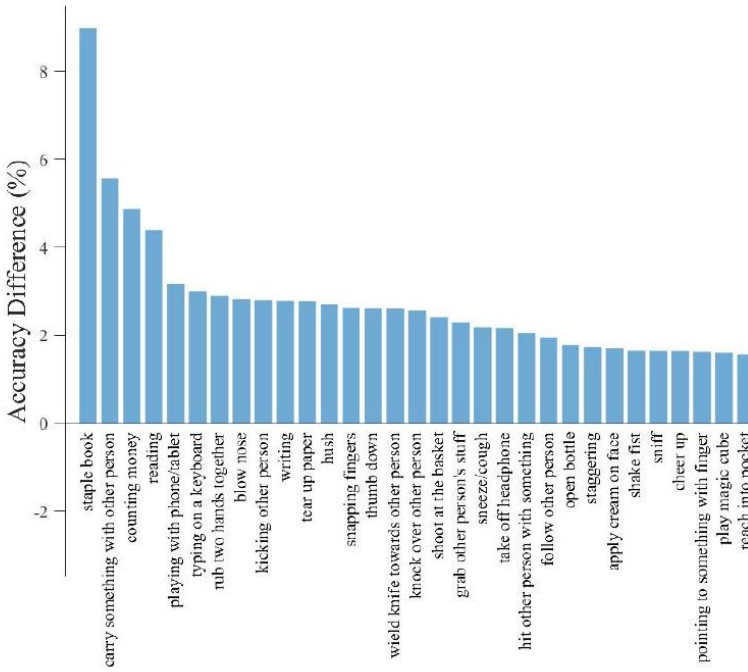

Figure 5. Action classes with accuracy difference higher than 1% between ProtoGCN and PYSKL on NTU-120 dataset. Most improved classes involve fine-grained movements.

5. Conclusion

We have presented ProtoGCN, a novel prototypical perspective for skeleton-based action recognition that addresses the fundamental challenge of distinguishing similar actions through fine-grained motion discrimination. By modeling action dynamics as combinations of reusable motion prototypes, our approach achieves superior performance across multiple challenging benchmarks.

The key insight underlying our work is that human actions can be naturally decomposed into fundamental motion patterns that can be learned and reused across different action categories. Our Prototype Reconstruction Network enables this decomposition by maintaining a learnable memory of prototypes and using attention-based addressing to reconstruct enhanced motion representations. The Motion Topology Enhancement module provides richer input by computing both intra-sample correlations and inter-sample distinctions, while the Class-Specific Contrastive Learning objective enforces class-level separability through momentum-updated prototypes.

Our experimental results demonstrate state-of-the-art performance on four challenging datasets: NTU RGB+D (93.8%/97.8%), NTU RGB+D 120 (90.9%/92.2%), Kinetics-Skeleton (51.9% Top-1), and FineGYM (95.9%). Importantly, our method shows the most significant improvements on the most challenging action classes, with over 3% gains on difficult categories. This validates that prototype reconstruction effectively captures subtle discriminative patterns that are crucial for fine-grained action recognition.

The comprehensive ablation studies confirm that all three components of ProtoGCN work synergistically, with the Prototype Reconstruction Network providing the largest contribution. The qualitative analysis further demonstrates that our method learns more discriminative joint relationships compared to existing approaches, particularly for actions involving fine-grained hand or foot movements.

While ProtoGCN advances the state-of-the-art in skeleton-based action recognition, several directions remain for future exploration. The prototype memory could be extended to hierarchical structures to capture multi-scale motion patterns. The attention mechanism could incorporate temporal dynamics for better sequential modeling. Additionally, the approach could be extended to multi-modal fusion with RGB data to leverage complementary information sources.

Our work establishes prototypical learning as a promising paradigm for skeleton-based action recognition, opening new research directions in compositional motion modeling and fine-grained action discrimination.

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